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PROBLEM 1:

Backtracking Search – Doctor Shift Assignment

Problem

Assign **3 doctors (D1, D2, D3)** to **3 shifts (Morning, Afternoon, Night)** satisfying all constraints.

Constraints

1. D1 cannot work **Night**.
2. D2 must work **before** D3.
3. Only one doctor per shift.
4. D3 cannot work **Morning**.
5. Every doctor gets exactly one shift.

Backtracking Steps

Step 1: Assign D1 = Morning ✓

- Constraint Check: D1 is not Night ✓

Current Assignment:

D1 → Morning

Step 2: Assign D2 = Afternoon ✓

Current Assignment:

D1 → Morning

D2 → Afternoon

Step 3: Assign D3 = Night ✓

Constraint Checks:

- D3 is not Morning ✓
- D2 before D3 (Afternoon < Night) ✓
- One doctor per shift ✓
- All doctors assigned ✓

✓ Solution Found

Search Tree

Start

D1 = Morning

D2 = Afternoon

D3 = Night

✓ Valid Solution

Final Schedule

Doctor Shift

D1 Morning

D2 Afternoon

D3 Night

Conclusion

Final valid assignment:

D1 → Morning, D2 → Afternoon, D3 → Night

This schedule satisfies **all the given constraints** and is obtained using the **Backtracking Search algorithm**.

2. BFS Search Algorithm for Robot Navigation

Given:

- Grid Size = 5×5
- Start = S
- Goal = G

Constraints:

1. Movement allowed: **Up, Down, Left, Right**
2. Cost of each move = **1**
3. Robot cannot pass through obstacles.
4. Find the shortest path using **BFS**.

Algorithm (BFS)

Step 1: Place the **Start node (S)** in the queue.

Queue: [S]

Step 2: Remove the first node from the queue and mark it as visited.

Step 3: Check whether it is the Goal (G).

- If **Yes**, stop.
- If **No**, expand its valid neighbours (Up, Down, Left, Right).

Step 4: Ignore:

- Obstacle cells
- Already visited cells

Step 5: Insert all valid neighbouring nodes into the queue.

Step 6: Repeat Steps 2–5 until the Goal node is reached.

Node Expansion & Queue Updates

Step	Expanded Node	Queue
1	S	[Adjacent Nodes]
2	First Adjacent Node	[Remaining Nodes]
3	Next Node	[New Adjacent Nodes]
4	Continue BFS	[...]
Final	G	Goal Reached

Optimal Path

$S \rightarrow \dots \rightarrow G$

(BFS always finds the **shortest path** when every move has the same cost.)

Path Cost

$$\boxed{\text{Total Cost} = \text{Number of Moves} \times 1}$$

Example:

If the shortest path has **8 moves**,

$$\text{Total Cost} = 8 \times 1 = 8$$

Conclusion

- BFS explores nodes **level by level**.
- Obstacles are **not traversed**.
- Since each move has the same cost, BFS guarantees the **optimal (shortest) path** from S to G

3. Autonomous Rescue Robot

a) Analysis of Constraints

Constraint	Effect on Robot's Decision
Move Up, Down, Left, Right	Robot searches only in four directions.
Obstacles	Robot avoids blocked cells and finds another path.
Limited sensing	Robot can observe only adjacent cells and explores step by step.
Move cost = 1	Robot tries to reduce the number of moves.
Risky zone cost = +2	Robot prefers safer paths with lower total cost.
Dynamic knowledge	Robot updates its map whenever it discovers new obstacles or paths.

b) Solution Approach (Online Search Agent + Uniform Cost Search)

Steps

1. Start at **S**.
2. Observe adjacent cells.
3. Add all safe and valid paths to the priority queue.
4. Select the path with the **lowest total cost**.
5. Move to the selected cell.
6. Update the map if new obstacles are detected.
7. Repeat until the robot reaches **G**.

Cost Formula

Total Cost = Movement Cost + Risk Cost

Movement Cost = **1 per step**

Risky Zone = **Additional 2**

c) AI Concepts

Intelligent Agent

- The robot senses the environment.
- It makes decisions.
- It performs actions to reach the survivor safely.

Rationality

- Chooses the path with the **minimum total cost**.

- Avoids obstacles and risky areas whenever possible.

Environment Type

- **Partially Observable:** Only nearby cells are visible.
- **Dynamic:** Environment information changes as the robot moves.
- **Sequential:** Current action affects future actions.
- **Discrete:** The robot moves from one grid cell to another.

d) Justification

Uniform Cost Search (UCS) with an Online Search Agent is the most suitable because:

- It always finds the **minimum-cost path**.
- It considers both **movement cost** and **risk cost**.
- It can **update decisions dynamically** when new obstacles are discovered.
- It is suitable for **partially observable disaster environments**.

Conclusion

The rescue robot intelligently explores the environment, updates its knowledge continuously, avoids obstacles and risky zones, and reaches the survivor using the **minimum-cost safe path**. This makes **Uniform Cost Search with an Online Search Agent** the best approach for the given problem.